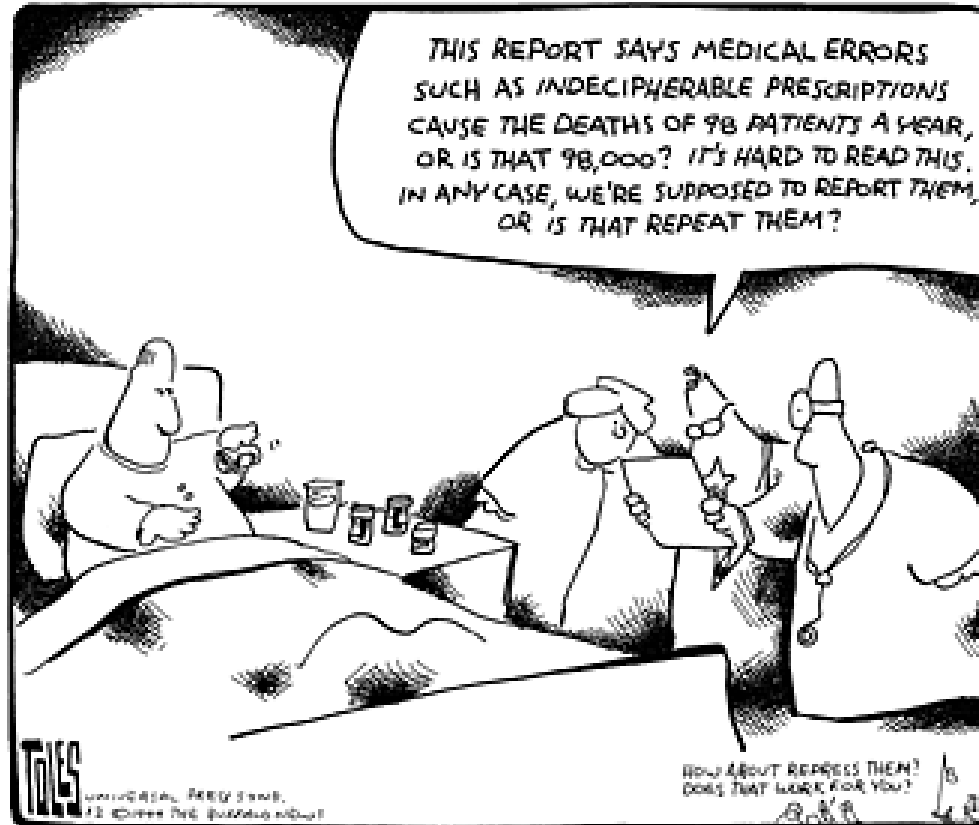




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Outline:

Basics

Most Common PDFs

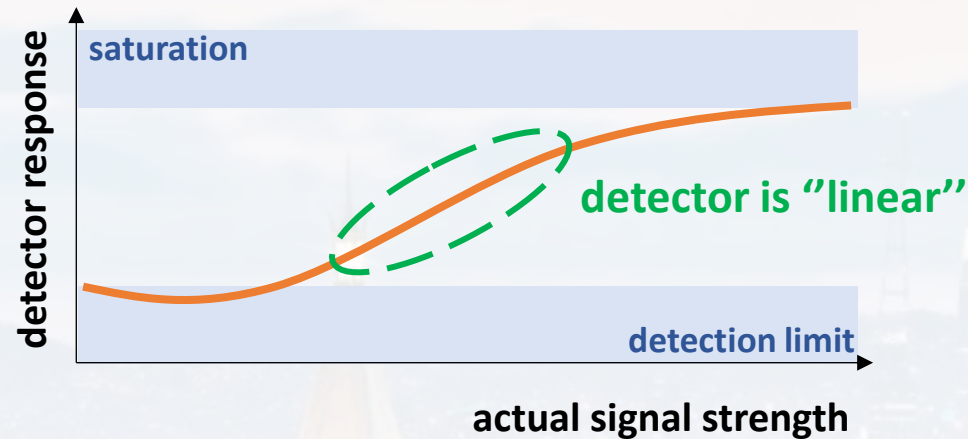
- uniform
- binomial
- Poissonian
- Normal/Gaussian

Error Estimation

Bayesian Statistics



- **systematic errors**: calibration, non-linearity of the detector

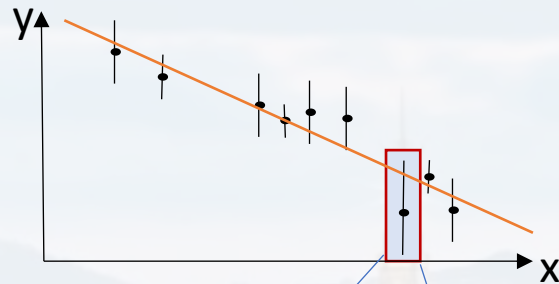


- **statistical errors**: limited precision, natural variance of the data
→ spread of the data around **an average value**



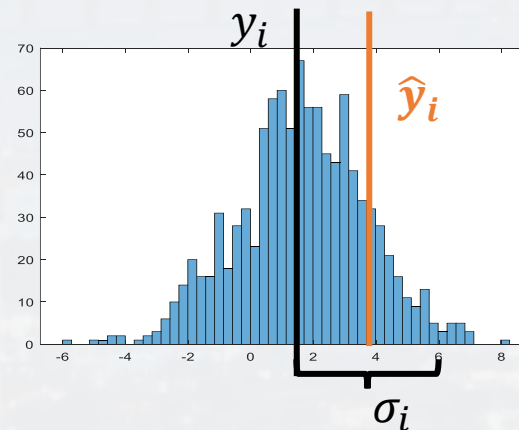
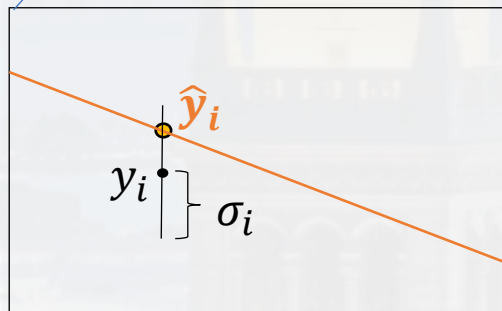
assumption: far from the detection limit and saturation:

the spread follows approximately a **normal distribution**.

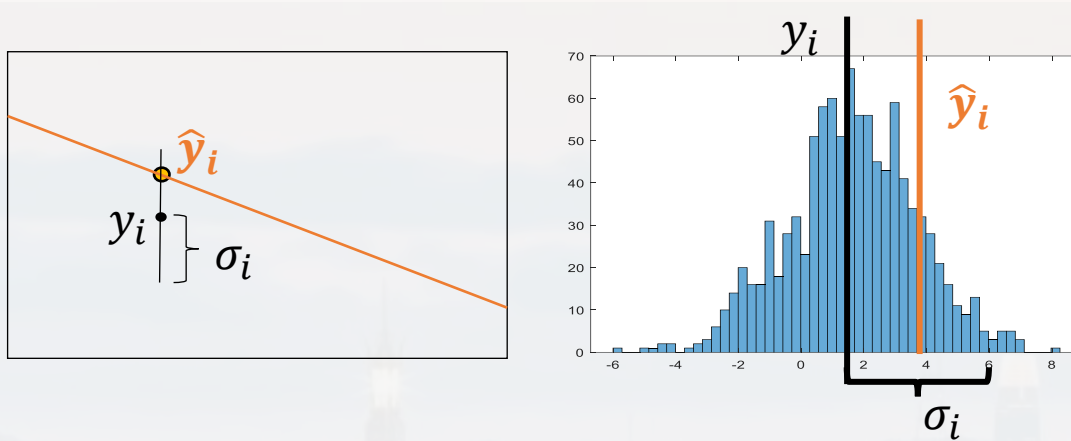


fitting a model ($\hat{y}_i$, orange line) to data points (x_i, y_i) each with an error bar σ_i

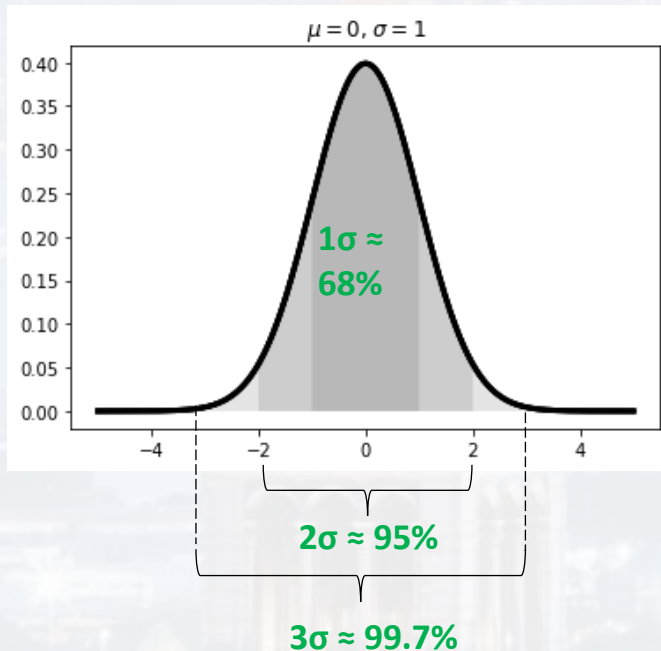
each data point x_i has been drawn from $N(\mu_i = y_i, \sigma_i)$



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$



for large ($> 50 \dots 100$) N (number of data points):

$\approx 2/3$ of the data points should be consistent with the model within their 1σ error bars

$\approx 95\%$ of the data points should be consistent with the model within their 2σ error bars

$\approx 99.7\%$ of the data points should be consistent with the model within their 3σ error bars



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

based on this model: reduced chi square

$$\chi_{red}^2 = \frac{1}{df} \sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{\sigma_i} \right)^2$$

$$df = N - p - 1$$

N : number of data points

p: number of fit parameter (model)

given a fitted model: χ_{red}^2 is a **measure of the fit quality!**



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

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$$df = N - p - 1$$

N : number of data points
p: number of fit parameter (model)

given a fitted model: χ_{red}^2 is a **measure of the fit quality!**

example

good fit: $y_i - \hat{y}_i$ should be within σ_i for 2/3 of all data points, see $N(\mu_i = y_i, \sigma_i)$

therefore $\frac{y_i - \hat{y}_i}{\sigma_i} \approx 1$

therefore $\sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{\sigma_i} \right)^2 \approx N$

N should be **much larger** than p, therefore $df \approx N$

hence, $\chi_{red}^2 \approx 1$



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

based on this model: reduced chi square

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$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

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≈ **2/3** of the data points should be consistent with the model within their **1σ** error bars

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≈ **99.7%** of the data points should be consistent with the model within their **3σ** error bars

$\chi_{red}^2 \approx$

1.0 excellent fit

1.0...1.5 acceptable fit

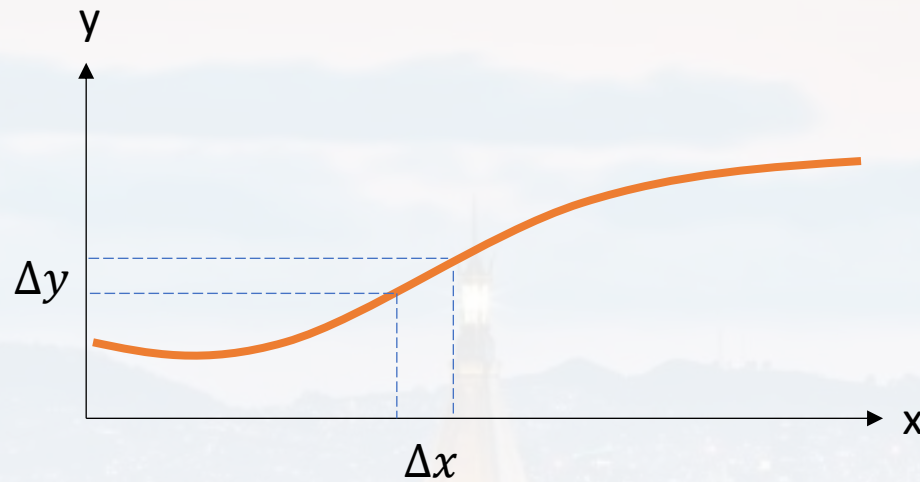
1.5...1.7 bad fit

>2.0 not acceptable

<<1.0 suspicious, errors are overestimated!



error propagation



$$\frac{dy}{dx} \approx \frac{\Delta y}{\Delta x}$$

for $\Delta x \ll x$

$$\Delta x \left| \frac{dy}{dx} \right| \approx \Delta y$$

example:

$$V = \frac{4}{3} \pi r^3$$

$\Delta V = ?$ given Δr

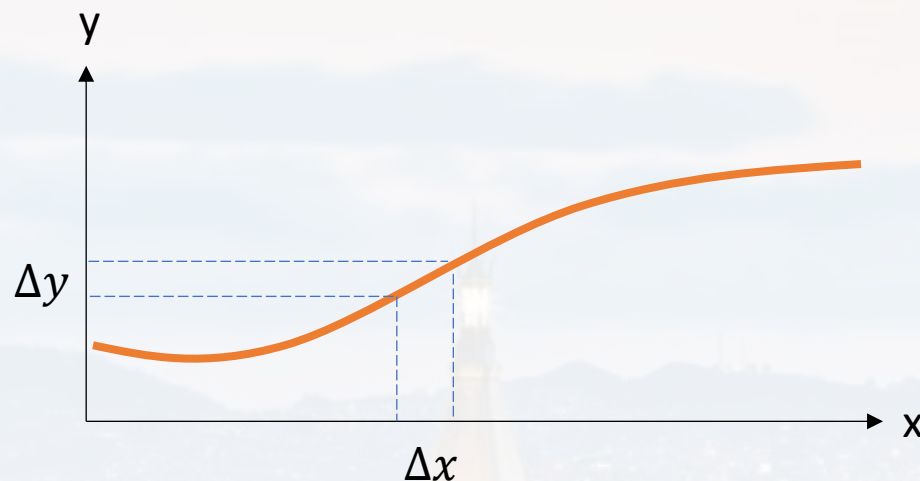
$$\Delta V = \frac{dV}{dr} \Delta r = 4 \pi r^2 \Delta r$$

$$\boxed{\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}}$$

$\Delta r, \Delta V \approx 1\sigma$



error propagation



$$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

$$\Delta r, \Delta V \approx 1\sigma$$

$$\Delta V = \frac{dV}{dr} \Delta r = 4 \pi r^2 \Delta r$$

$$\begin{array}{ll} \text{radius: } r = & 1.0 \mu\text{m} \\ \Delta r = & 0.1 \mu\text{m} \end{array}$$

$$V = \frac{4}{3} \pi (1.0 \mu\text{m})^3 = 4.18879020 \dots \mu\text{m}^3$$

$$\begin{aligned} \Delta V &= 4 \pi r^2 \Delta r = 4 \pi (1.0 \mu\text{m})^2 0.1 \mu\text{m} \\ &= 1.2566370614359172 \mu\text{m}^3 \end{aligned}$$

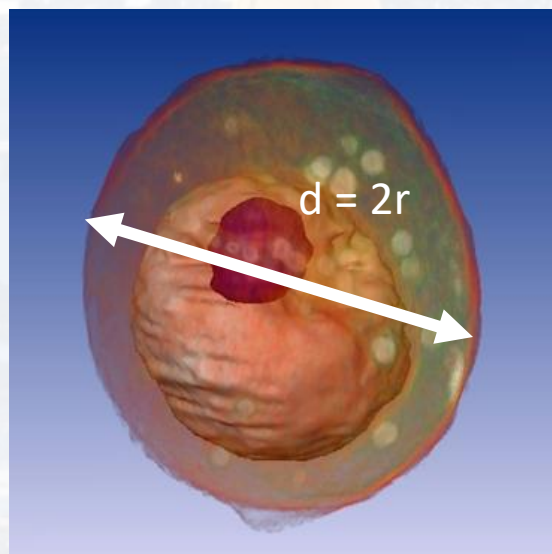
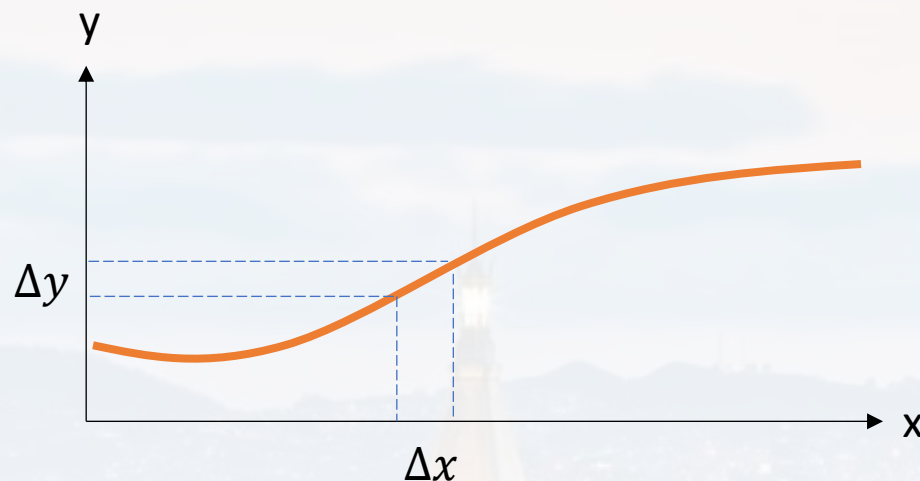


Image: NIH



error propagation



$$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

$$\Delta r, \Delta V \approx 1\sigma$$

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radius: $r = 1.0 \mu\text{m}$
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$$V = \frac{4}{3} \pi (1.0 \mu\text{m})^3 = \mathbf{4.18879020} \dots \mu\text{m}^3$$

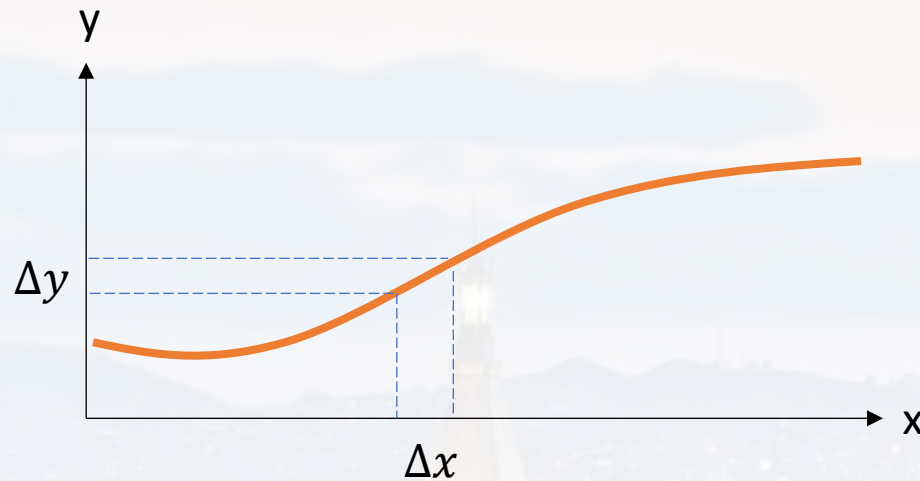
$$V = (4.2 \pm 1.3) \mu\text{m}^3$$

$$\Delta V = 4 \pi r^2 \Delta r = 4 \pi (1.0 \mu\text{m})^2 0.1 \mu\text{m}$$

$$= \mathbf{1.2566370614359172} \mu\text{m}^3$$



error propagation



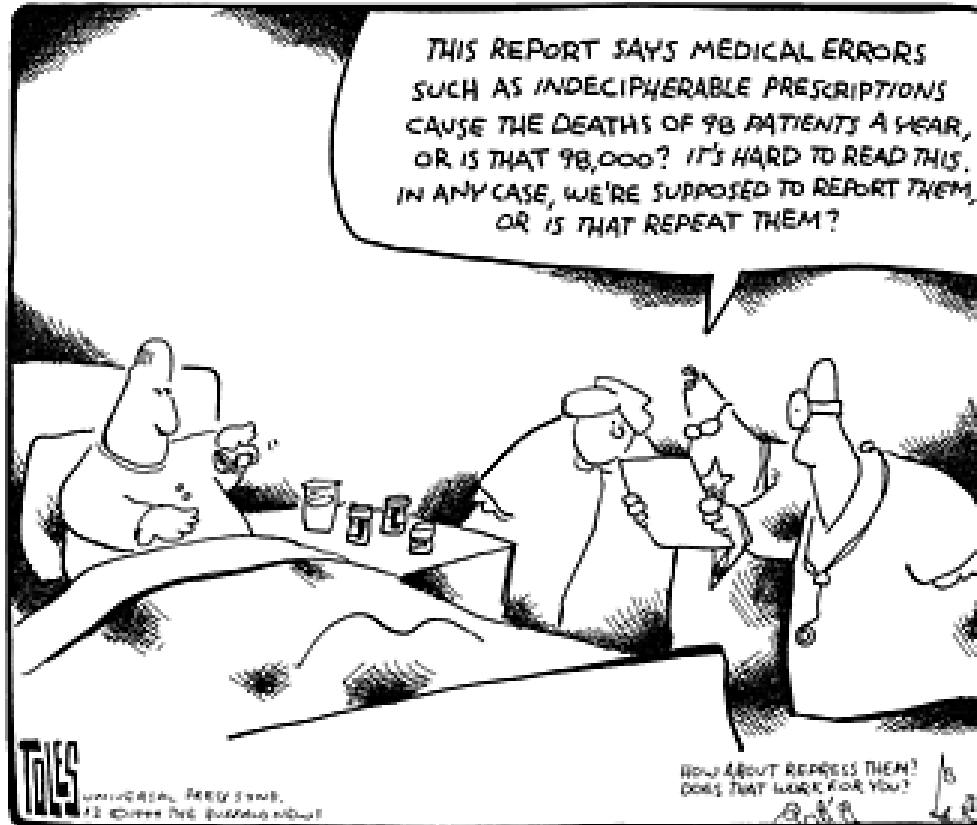
general:

$$\Delta f(max) = \sum_{i=1}^I \left| \frac{\partial f}{\partial x_i} \right| \Delta x_i \quad \text{maximum error estimation}$$

if x_i do **not correlate**, i. e. are **mutually independent**:

$$\Delta f^2 = \sum_{i=1}^I \left| \frac{\partial f}{\partial x_i} \right|^2 (\Delta x_i)^2$$

Note: $\Delta f(max)^2 > \Delta f^2$ because of the mixed terms $\left| \frac{\partial f}{\partial x_i} \right| \left| \frac{\partial f}{\partial x_j} \right| \Delta x_i \Delta x_j$ in $\Delta f(max)^2$



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Outline:

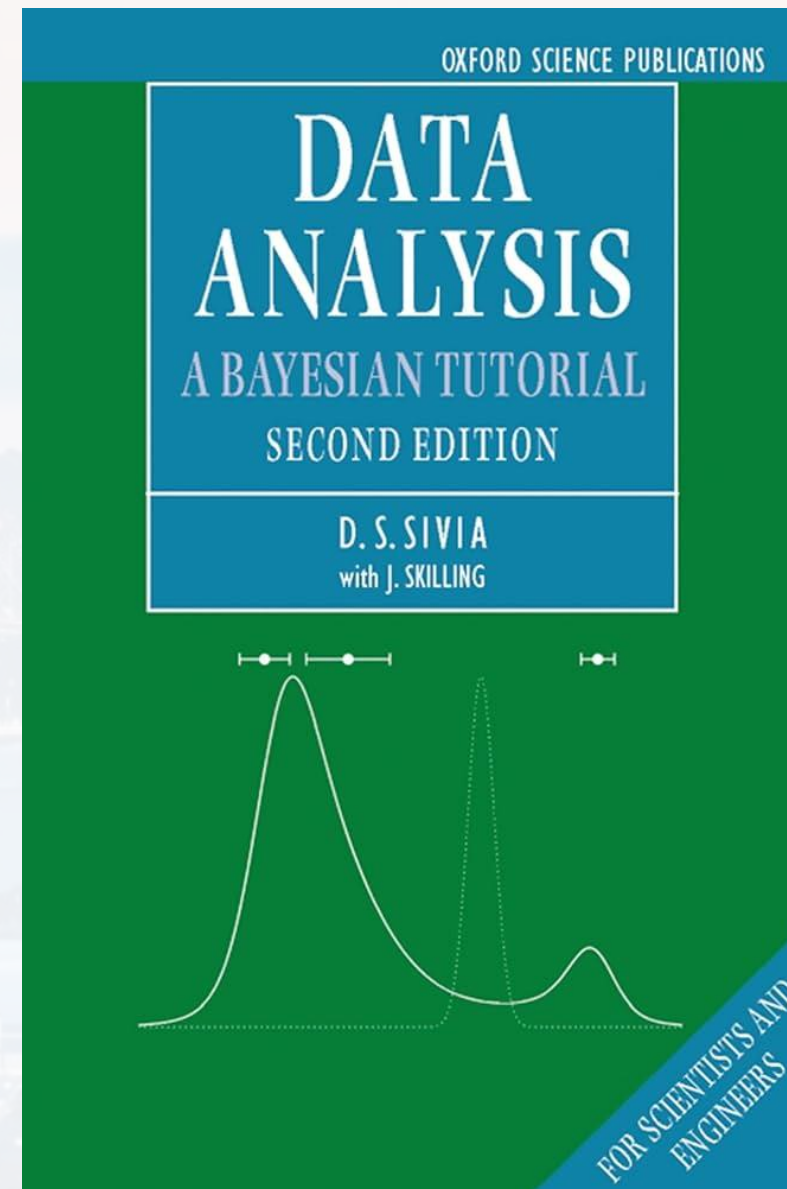
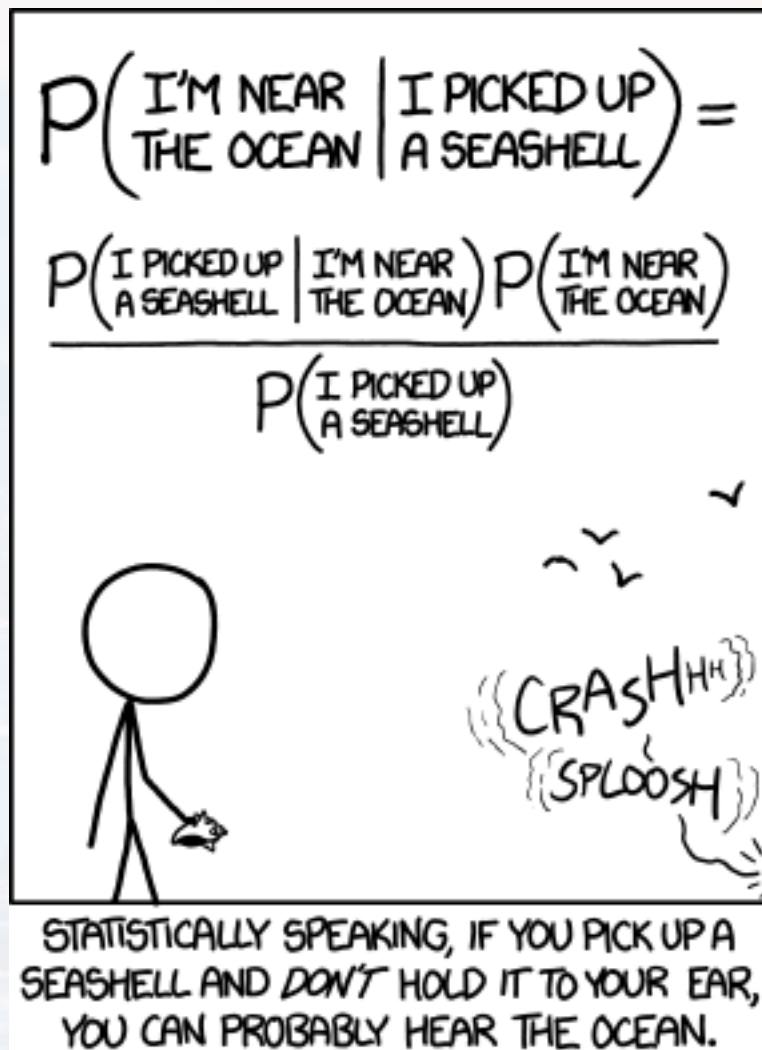
Basics

Most Common PDFs

- uniform
- binomial
- Poissonian
- Normal/Gaussian

Error Estimation

Bayesian Statistics





$P(A \cap B)$ probability **P** that the events **A** and **B** occur

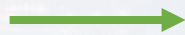
so far: A and B were independent $P(A \cap B) = P(A)P(B) = P(B)P(A)$

now: **conditional probabilities** | “given” or “under the condition”



Thomas Bayes
(1701 - 1761)

$$\begin{aligned} P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A) \end{aligned}$$



$$P(A|B)P(B) = P(B|A)P(A)$$

Bayes Theorem

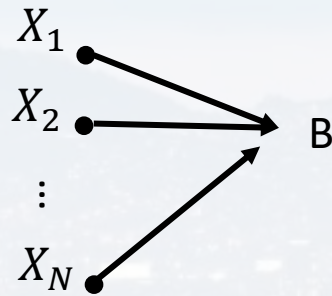
$$\text{posterior } \mathbf{P(A|B)} = \frac{P(B|A)\mathbf{P(A)}}{P(B)} \text{ prior}$$



$$P(A|B)P(B) = P(B|A)P(A)$$

Bayes Theorem

posterior $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$ prior



$$P(B) = \sum_{n=1}^N P(B|X_n)P(X_n)$$

$$P(B) = \int P(B|X)P(X) dX$$

marginalization



Thomas Bayes
(1701 - 1761)

Probability $P(B)$ that I am going to be too late for a meeting:

$$P(B) = P(B|I \text{ forgot that I have a meeting}) P(I \text{ forgot that I have a meeting}) + \\ P(B|I \text{ got sick}) P(I \text{ got sick}) + \\ P(B|BART \text{ was too late}) P(BART \text{ was too late}) + \dots$$



example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Marginalization

$$P(B) = \sum_{n=1}^N P(B|X_n)P(X_n)$$

statement 1: If a person is **diseased**, there is a **95% probability** that the test is **positive**.

statement 2: The **prevalence** for the disease in the average **population** is **0.001%**.

statement 3: **5% of healthy** patients have **a positive result** (aka p-value).

A person takes the test and gets a positive test result. **What is the probability that the person is sick?**

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)} = \frac{\overset{\text{statement 1}}{0.95} P(D)}{P(+)} = \frac{\overset{\text{statement 2}}{0.95 \cdot 0.00001}}{P(+)} = \frac{\overset{\text{marginalization}}{0.95 \cdot 0.00001}}{P(+|D)P(D) + P(+|H)P(H)}$$



example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

Bayes Theorem

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$$\begin{aligned}
 P(D|+) &= \frac{P(+|D)P(D)}{P(+)} = \frac{\text{0.95 } P(D)}{P(+)} = \frac{\text{0.95} \cdot \text{0.00001}}{P(+)} = \frac{\text{0.95} \cdot \text{0.00001}}{P(+|D)P(D) + P(+|H)P(H)} \\
 &\quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \text{marginalization} \\
 &\quad \text{statement 1} \quad \text{statement 2} \\
 &= \frac{\text{0.95} \cdot \text{0.00001}}{P(+|D)P(D) + P(+|H)[\text{1} - P(D)]} \quad \text{complement probability}
 \end{aligned}$$



example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Marginalization

$$P(B) = \sum_{n=1}^N P(B|X_n)P(X_n)$$

statement 1: If a person is **diseased**, there is a **95% probability** that the test is **positive**.
statement 2: The **prevalence** for the disease in the average **population** is **0.001%**.
statement 3: **5% of healthy** patients have **a positive result** (aka p-value).

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|H)[1 - P(D)]}$$

$$= \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}} = \frac{1}{1 + \frac{0.05 [1 - 0.00001]}{0.95 \cdot 0.00001}} = 1/5000$$



example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

statement 1:

sensitivity

$P(D|+) = 95\%$

statement 2:

prior

$P(D) = 0.001\%$

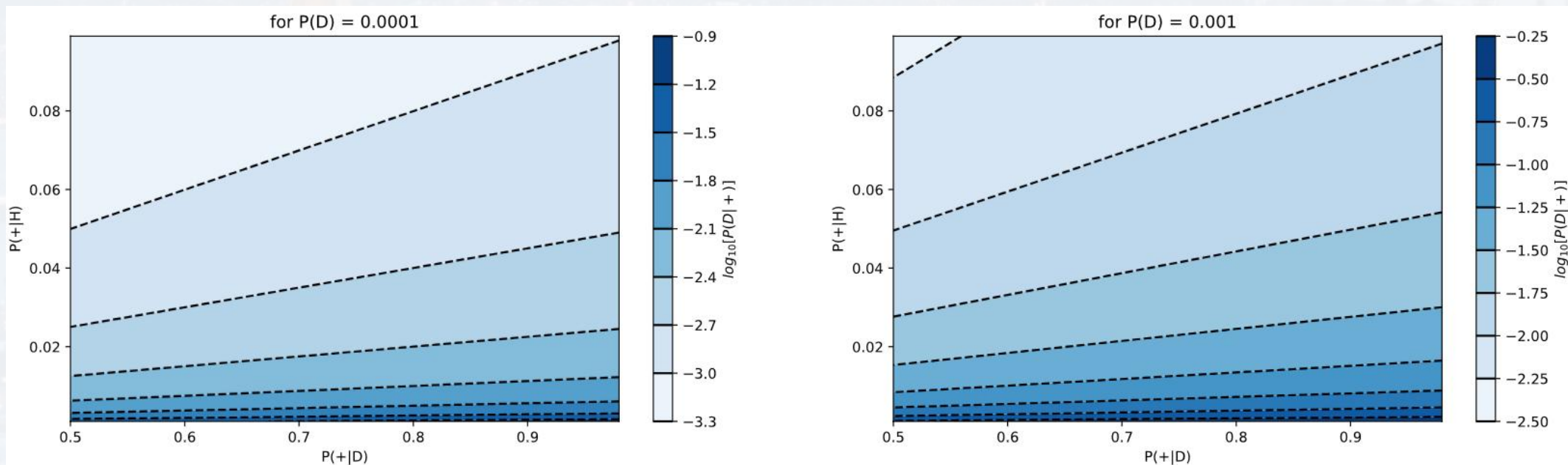
statement 3:

p-value or false positive rate

$P(+|H) = 5\%$

$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

check: `PlotPD_Plus.py`





example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

statement 1:

sensitivity

$P(D|+) = 95\%$

statement 2:

prior

$P(D) = 0.001\%$

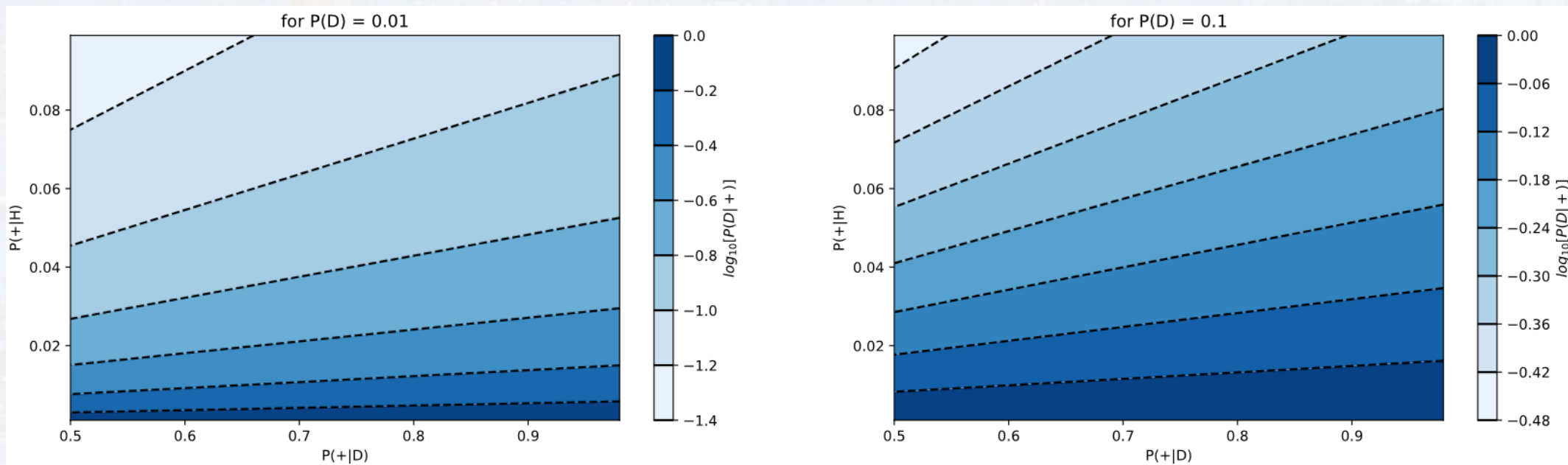
statement 3:

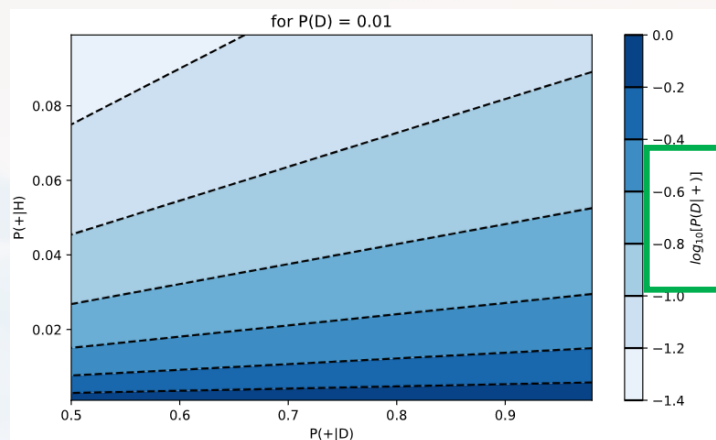
p-value or false positive rate

$P(+|H) = 5\%$

$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

check: `PlotPD_Plus.py`





statement 1:

sensitivity

$P(D|+) = 95\%$

statement 2:

prior

$P(D) = 0.001\%$

statement 3:

p-value or false positive rate

$P(+|H) = 5\%$

odds ratios:

$$\rho_1 = \frac{P(+|H)}{P(+|D)}$$

$$\rho_2 = \frac{1 - P(D)}{P(D)}$$

$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

log odds ratios: $r_1 = \log \left[\frac{P(+|H)}{P(+|D)} \right]$

$$r_2 = \log \left[\frac{1 - P(D)}{P(D)} \right]$$

$$P(D|+) = \frac{1}{1 + e^{r_1} e^{r_2}}$$



log odds ratios: $r_1 = \log \left[\frac{P(+|H)}{P(+|D)} \right]$

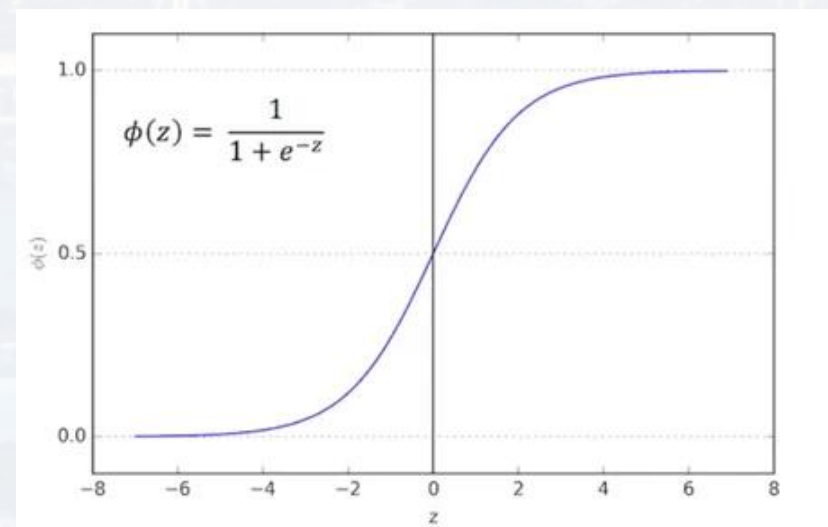
$$r_2 = \log \left[\frac{1 - P(D)}{P(D)} \right]$$

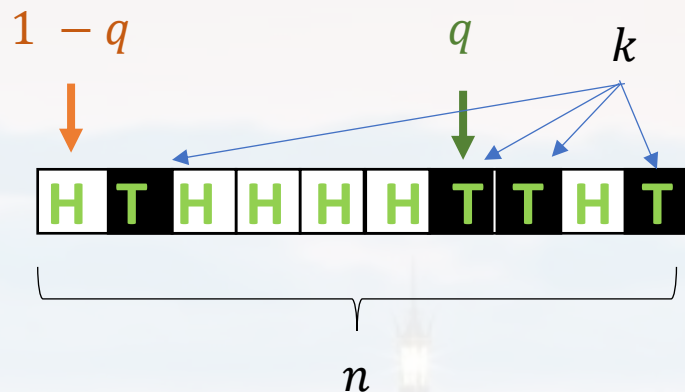
$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

$$P(D|+) = \frac{1}{1 + e^{r_1}e^{r_2}}$$

logistic (or logit or sigmoid) function

- logistic regression
- transfer function ANN
- bound growth (Verhulst equation)
- binding affinity ligand/receptor





probability of having a sequence of k tails and $n-k$ heads

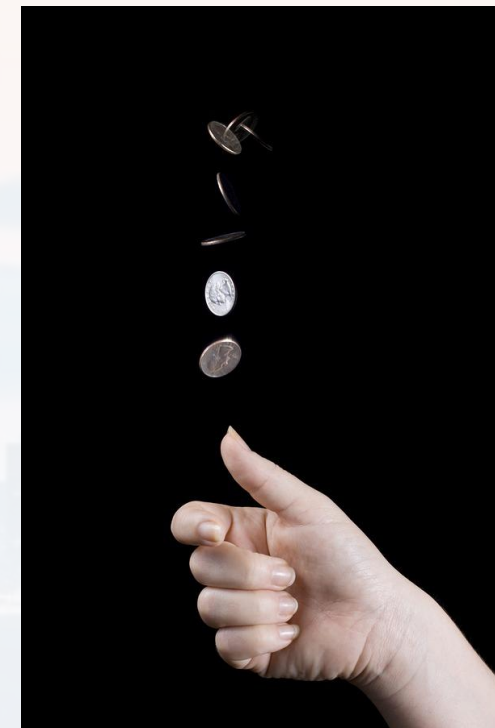
$$p_{tot} = \prod_i q_i^{n_i} = q^k (1 - q)^{n-k}$$

probability of having **any** sequence of k tails and $n-k$ heads

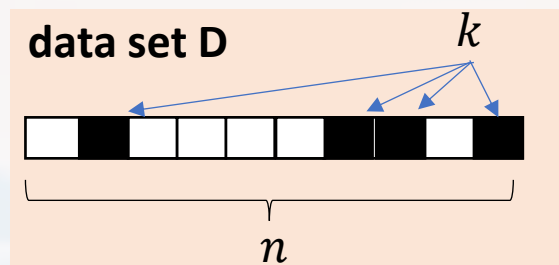
$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

$$\frac{n!}{k!(n-k)!} =: \binom{n}{k} \quad \text{``n choose k''}$$



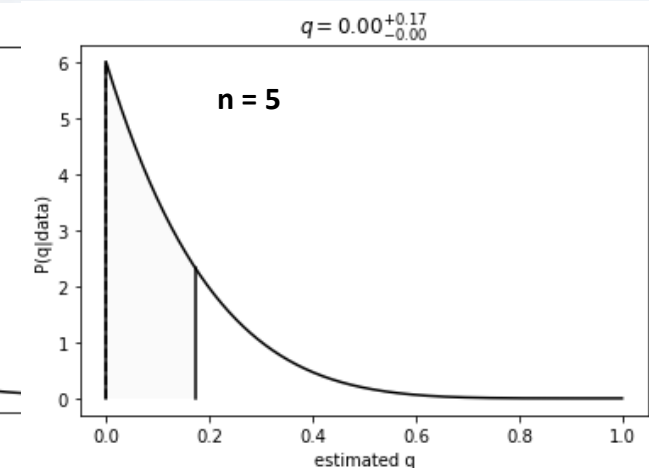
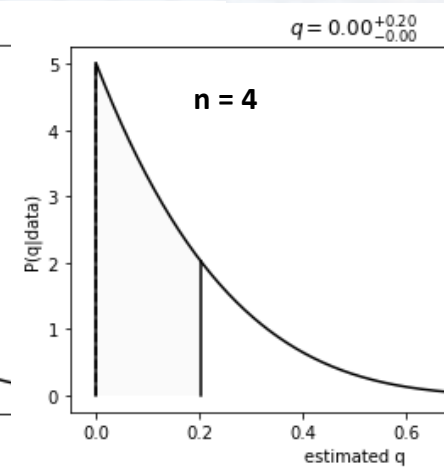
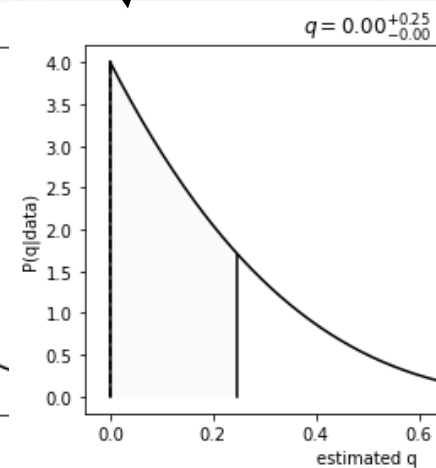
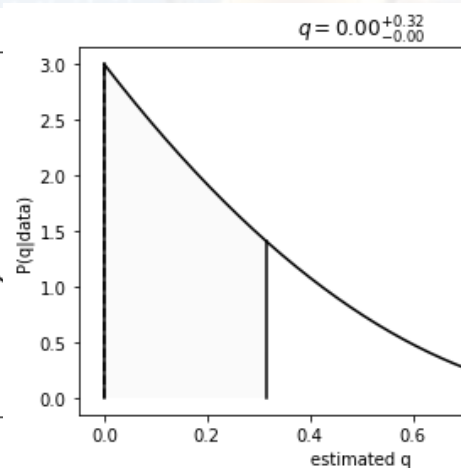
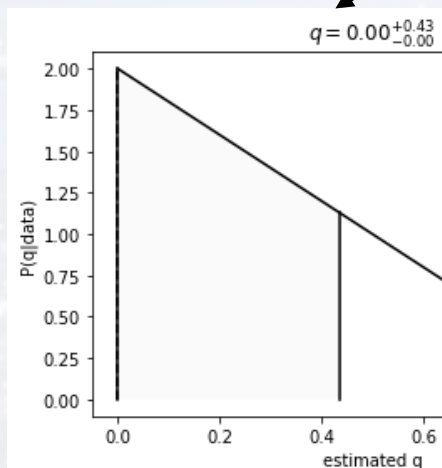
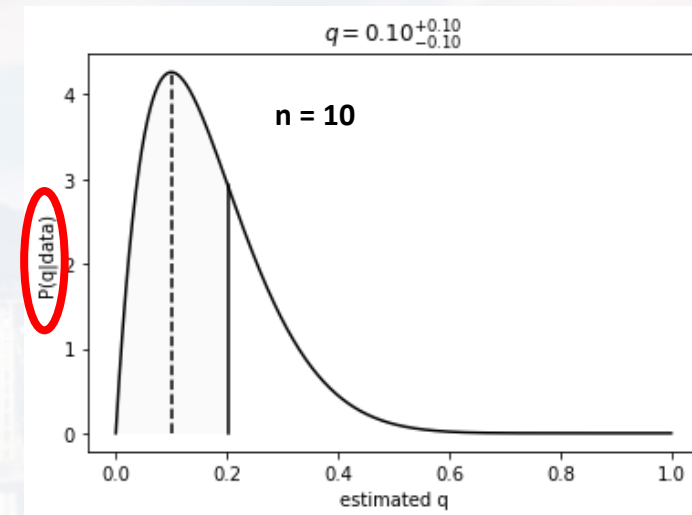
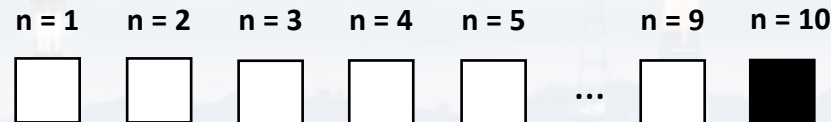
fair coin? $q = 0.5$???

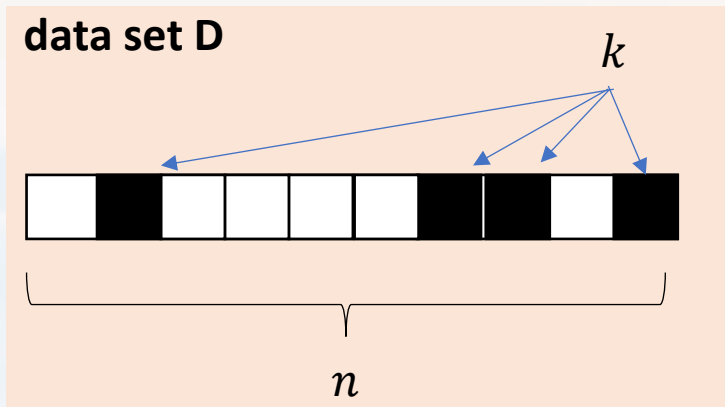


$$q = ?$$

goal:

- $P(q|D)$ **Parameter Estimation**
- the larger D , the more certain q
→ learning





$q = ?$

goal:

- $P(q|D)$
- the larger D , the more certain q
→ learning

$$P(k|n, q) = \binom{n}{k} q^k (1 - q)^{n-k}$$

Bayes' theorem:

likelihood function (here: binomial)

$$P(q|\text{data set}) = \frac{P(\text{data set}|q)P(q)}{P(\text{data set})}$$

prior

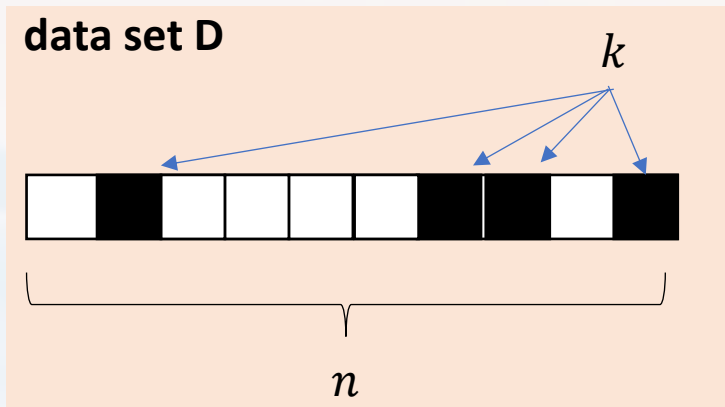
evidence (const wrt q)

$$= \frac{\binom{n}{k} q^k (1-q)^{n-k}}{P(D)} P(q)$$

$$\sim q^k (1 - q)^{n-k} P(q)$$

$P(D)$ and $\binom{n}{k}$ are no functions of q





$q = ?$

goal:

- $P(q|D)$
- the larger D , the more certain q
→ learning

$$P(k|n, q) = \binom{n}{k} q^k (1 - q)^{n-k}$$

$$P(q|data\ set) = \frac{P(data\ set|q)P(q)}{P(data\ set)}$$

$$= \frac{\binom{n}{k} q^k (1-q)^{n-k}}{P(D)} P(q)$$

$$\sim q^k (1 - q)^{n-k} P(q)$$

$$\sim q^k (1 - q)^{n-k}$$

max. entropy: $P(q) = \text{const}$
if no prior information about q

$$P(q|data\ set) = \frac{q^k (1 - q)^{n-k}}{\int_0^1 q^k (1 - q)^{n-k} dq}$$



check out `bayesian_bino.py`

```
n1 = 4
```

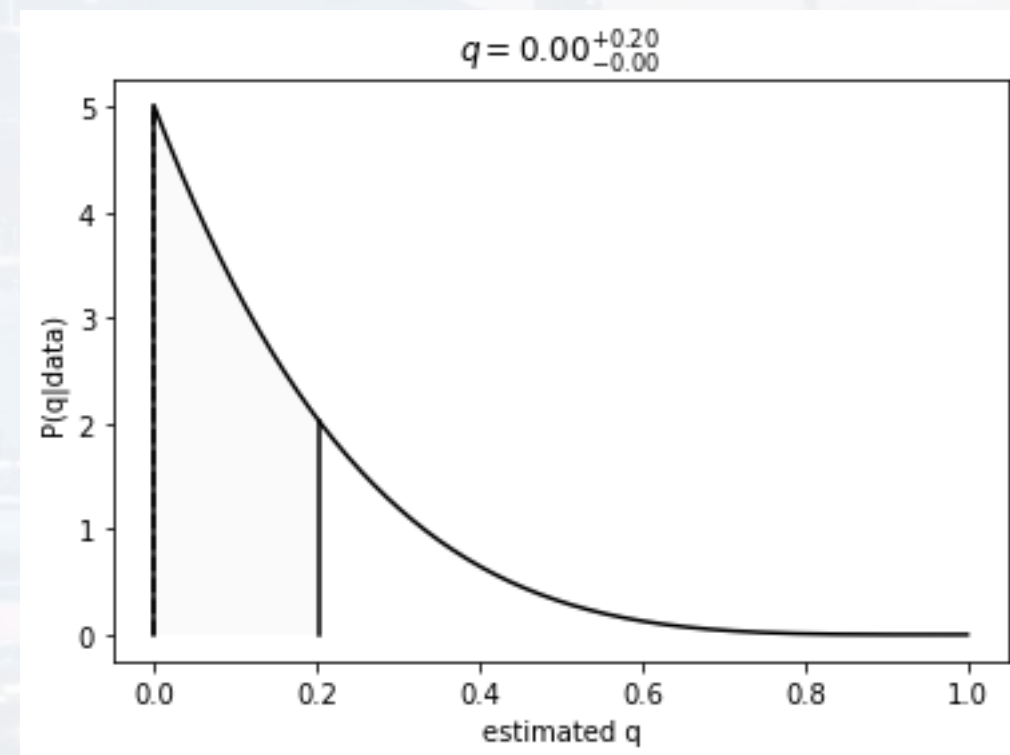
```
k1 = np.random.binomial(n1, 0.25)
```

creating artificial data set

note: in reality q is unknown!

```
[q1, b, _] = bayesian_bino(n1, k1)
```

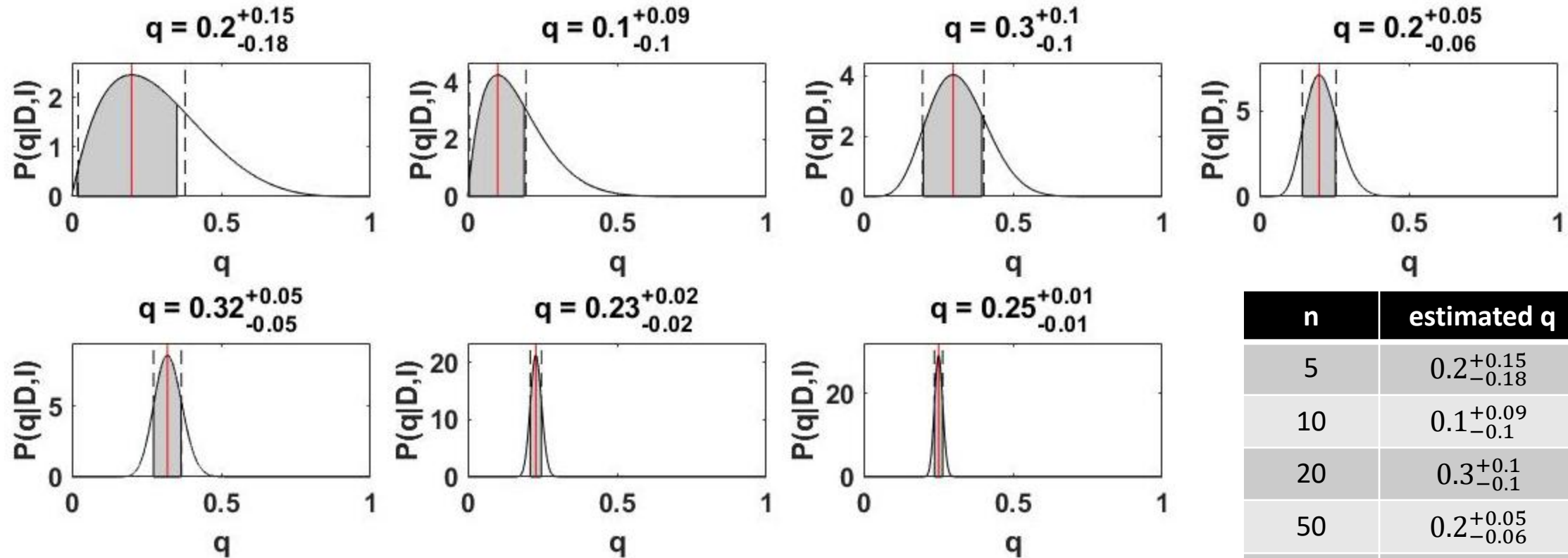
$$P(q|data\ set) = \frac{q^k(1-q)^{n-k}}{\int_0^1 q^k(1-q)^{n-k} dq}$$





check out `bayesian_bino.py`

$$P(q|data\ set) = \frac{q^k(1-q)^{n-k}}{\int_0^1 q^k(1-q)^{n-k} dq}$$

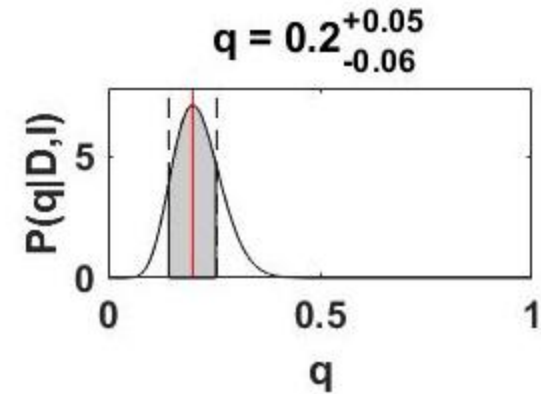
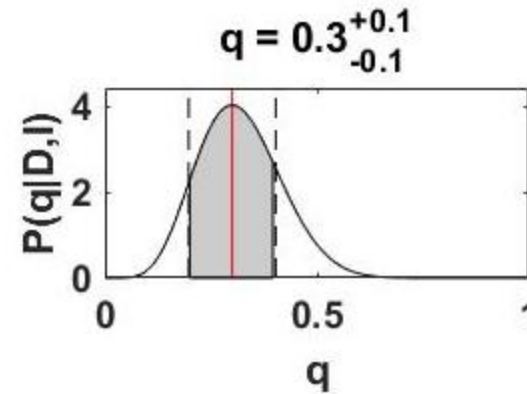
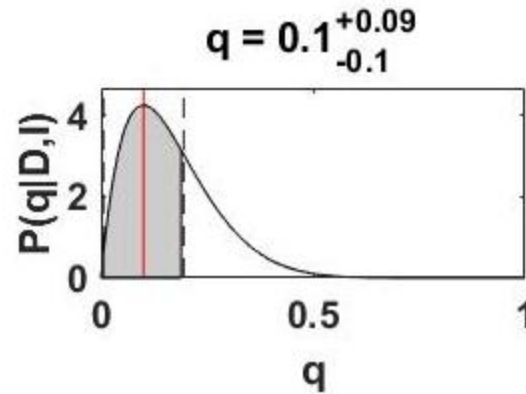
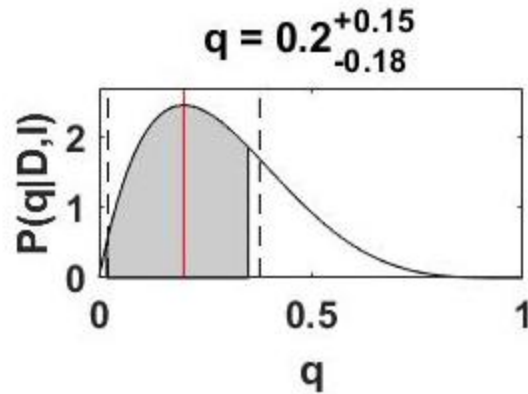


n	estimated q
5	$0.2^{+0.15}_{-0.18}$
10	$0.1^{+0.09}_{-0.1}$
20	$0.3^{+0.1}_{-0.1}$
50	$0.2^{+0.05}_{-0.06}$
100	$0.32^{+0.05}_{-0.05}$
500	$0.23^{+0.02}_{-0.02}$
1,000	$0.25^{+0.01}_{-0.01}$
infinity	0.25



check out `bayesian_bino.py`

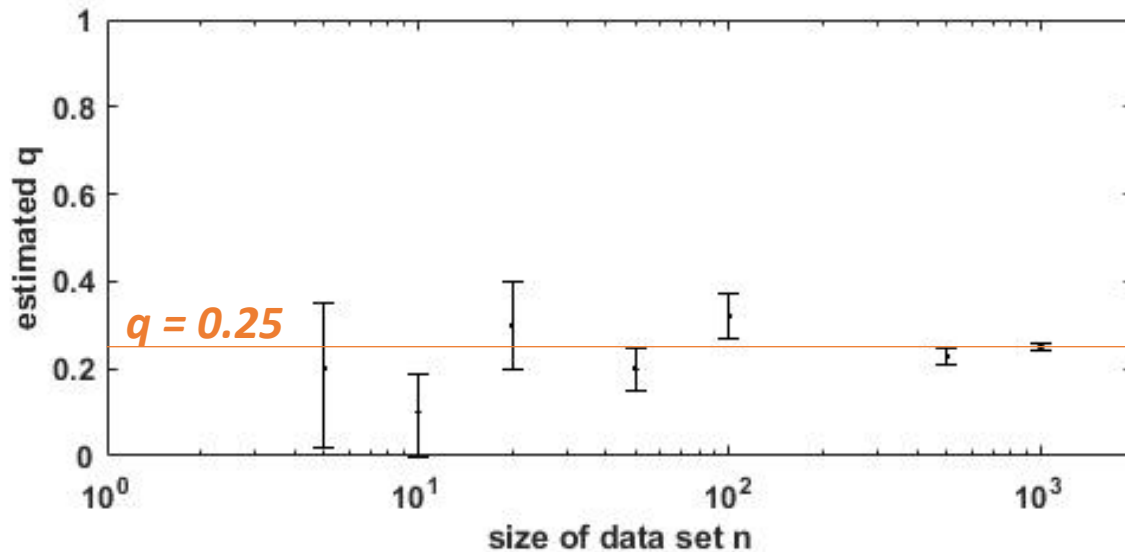
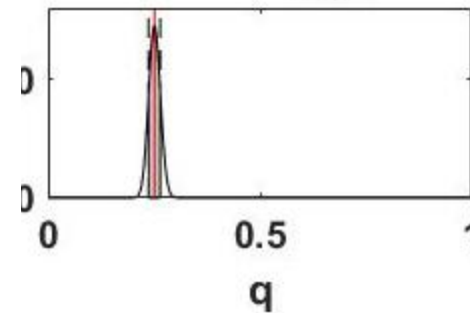
$$P(q|data\ set) = \frac{q^k(1-q)^{n-k}}{\int_0^1 q^k(1-q)^{n-k} dq}$$



$q = 0.32^{+0.05}_{-0.05}$

$q = 0.23^{+0.02}_{-0.02}$

$q = 0.25^{+0.01}_{-0.01}$



n	estimated q
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Of course, Bayesian Parameter Estimation works with **any other pdf**

goal:

- $P(q|D)$
- the larger D , the more certain q
→ learning

likelihood function

$$P(q|data\ set) = \frac{P(data\ set|q)P(q)}{P(data\ set)}$$

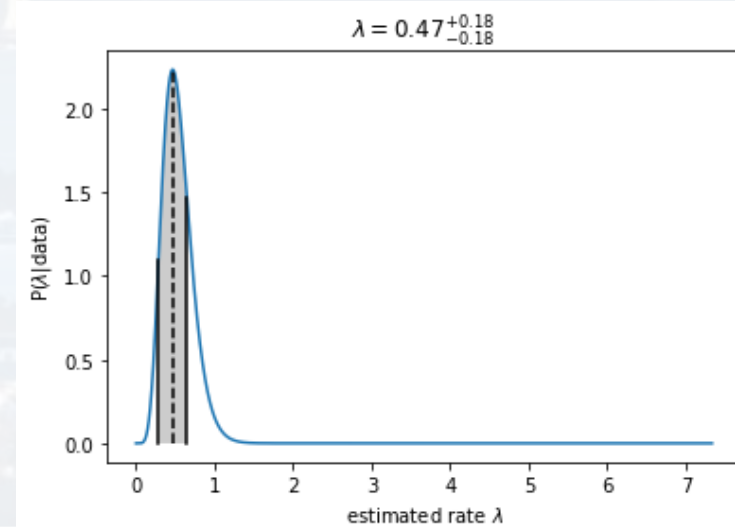
prior evidence (const wrt q)

What is the average number of WhatsUp messages I get every day?

Mon:	5	event	- has no duration
Tue:	7		- is rare
Wed:	1		
Thu:	3		→ Poissonian
Fri:	9		
Sat:	2		
Sun:	5		

```
data = np.random.poisson(lam = 0.4, 15)
poissfit(data)
```

$$P(k|\lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$





Of course, Bayesian Parameter Estimation works with **any other pdf**

goal:

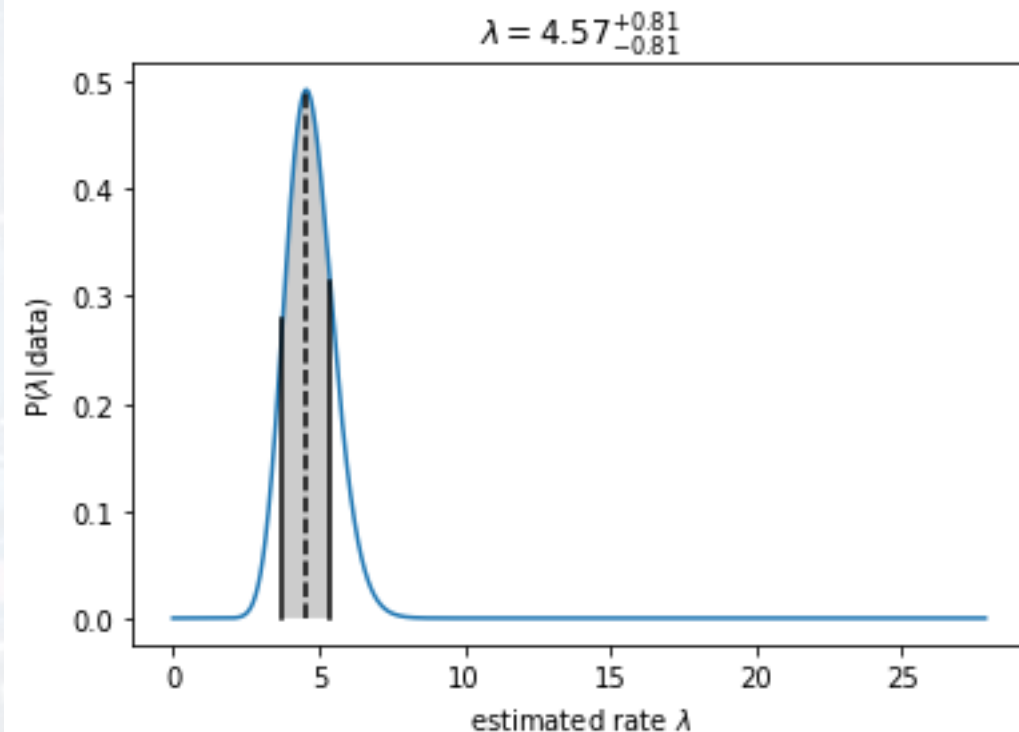
- $P(\mathbf{q}|\mathbf{D})$
- the larger $\mathbf{D}$, the more certain $\mathbf{q}$
→ learning

What is the average number of WhatsUp messages I get every day?

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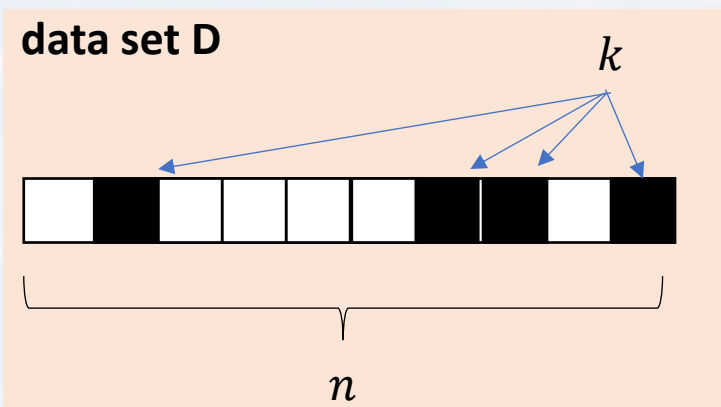
$$P(k|\lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

```
poissfit([5, 7, 1, 3, 9, 2, 5])
```



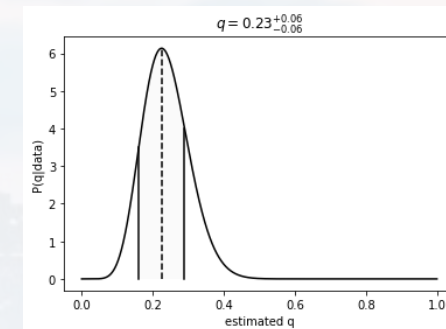


What if there is new data?



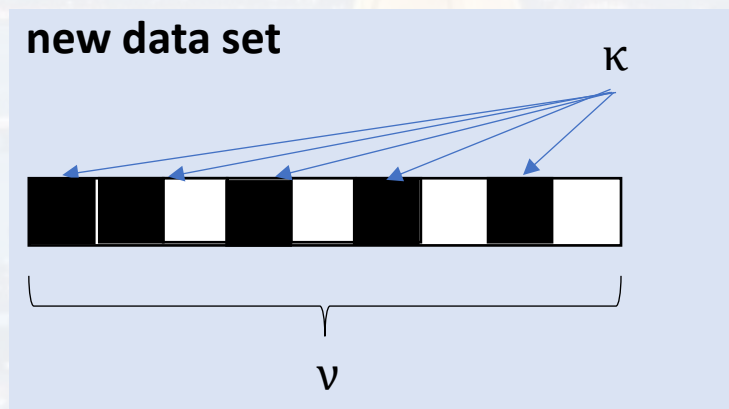
~~$q = ?$~~

$$P(q|\text{data set}) = \frac{q^k(1-q)^{n-k}}{\int_0^1 q^k(1-q)^{n-k} dq}$$



if there is prior information I about q :

$$P(q|\text{new data set}, I) = \frac{P(\text{new data set}|q, I) P(q, I)}{P(\text{new data set})}$$





What if there is new data?

$$P(q|\text{new data set}, I) = \frac{P(\text{new data set}|q, I) \mathbf{P(q, I)}}{P(\text{new data set})}$$

$$P(q|\text{data set}) = \frac{q^k(1-q)^{n-k}}{\int_0^1 q^k(1-q)^{n-k} dq}$$

$$= \frac{q^\kappa(1-q)^{\nu-\kappa} q^k(1-q)^{n-k}}{\int_0^1 q^\kappa(1-q)^{\nu-\kappa} q^k(1-q)^{n-k} dq}$$

$$= \frac{q^{k+\kappa}(1-q)^{\nu-\kappa+n-k}}{\int_0^1 q^{k+\kappa}(1-q)^{\nu-\kappa+n-k} dq}$$

$$= \int_0^1 q^{k+\alpha-1}(1-q)^{n-k+\beta-1} dq$$

often: $\kappa = \alpha - 1$
 $\beta = \nu - \kappa - 1$

Beta function

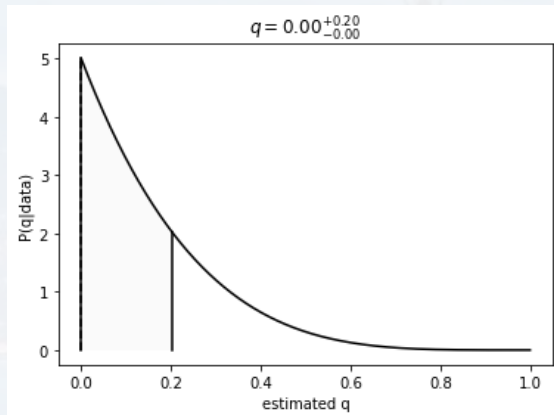


What if there is new data?

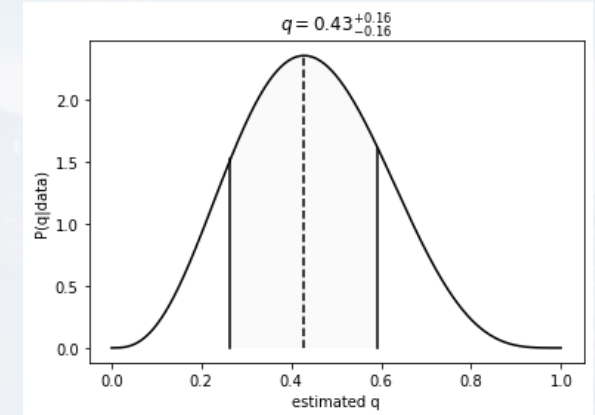
$$P(q|\text{new data set}, I) = \frac{q^\kappa(1-q)^{\nu-\kappa}}{\int_0^1 q^\kappa(1-q)^{\nu-\kappa} dq} \frac{q^k(1-q)^{n-k}}{q^k(1-q)^{n-k}}$$

```
n1 = 4
k1 = np.random.binomial(n1, q = 0.2)
[_, _, Prior] = bayesian_bino(n1, k1)
```

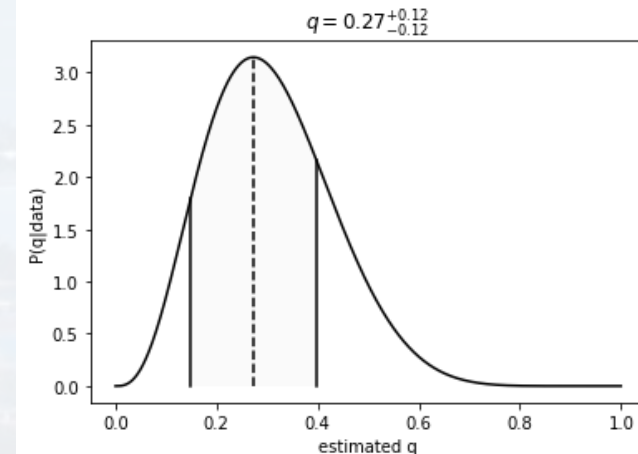
```
n2 = 7
k2 = np.random.binomial(n2, q = 0.2)
[_, _, _] = bayesian_bino(n2, k2)
```



$$P(q, I) = \frac{q^k(1-q)^{n-k}}{\int_0^1 q^k(1-q)^{n-k} dq}$$



```
[_, _, _] = bayesian_bino(n2, k2, Prior = Prior)
```

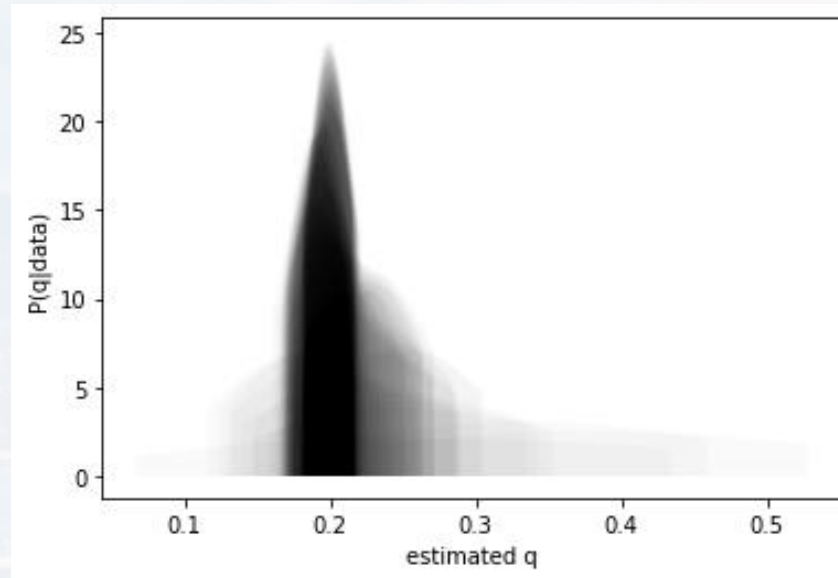




What if there is new data?

$$P(q|\text{new data set}, I) = \frac{q^\kappa(1-q)^{v-\kappa}}{\int_0^1 q^\kappa(1-q)^{v-\kappa}} \frac{q^k(1-q)^{n-k}}{q^k(1-q)^{n-k}} dq$$

The **posterior** from the **previous** experiment is the **prior** of the **next** experiment



→ we become more certain about the model parameters
→ learning!

→ see e.g. **Variational Auto Encoders**

2D images → 3D objects



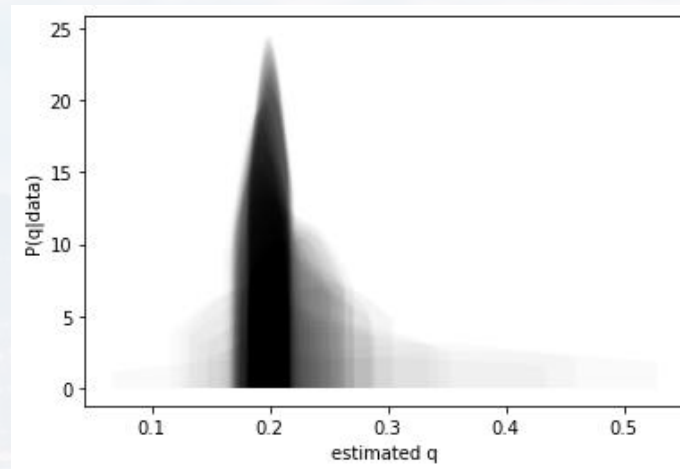
credit: StableAI



What if there is new data?

$$P(q|\text{new data set}, I) = \frac{q^\kappa(1-q)^{v-\kappa}}{\int_0^1 q^\kappa(1-q)^{v-\kappa}} \frac{q^k(1-q)^{n-k}}{q^k(1-q)^{n-k}} dq$$

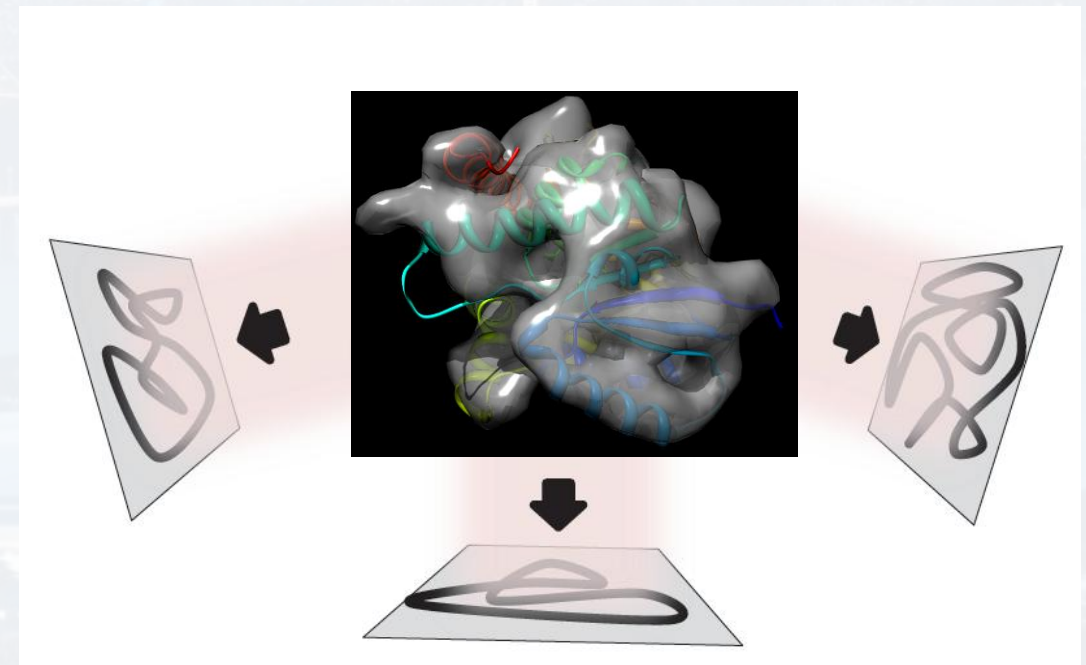
The **posterior** from the **previous** experiment is the **prior** of the **next** experiment



- we become more certain about the model parameters
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→ see e.g. **Variational Auto Encoders** 2D images → 3D objects

Cryo – EM: 3D structure from 2D images/projections



(image courtesy: Thomas Becker, GC LMU Munich)



Thank you very much for your attention!

